# ROJIN SAFAVI

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rojin-safavi
Ph.D. in Bioinformatics
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## AREAS OF EXPERTISE

Cell-Free DNA (cfDNA), Tumor Fraction Estimation, Minimal Residual Disease (MRD), Early Cancer Detection, Epigenomics (DNA Methylation), Variant Interpretation, Multi-Omic Integration, Biomarker Discovery, Clinical Assay Development, Analytical Validation, Next-Generation Sequencing (NGS), Machine Learning & AI, Statistical Modeling, Real-World Evidence (RWE), Electronic Health Records (EHR), CLIA/CAP Regulatory Compliance

## TECHNICAL SKILLS

**Programming Languages:** Python, R, Bash
**Machine Learning:** Supervised and unsupervised models, Clustering & Classification, Deep Learning (Transformers, VAEs, Multi-Modal Embedding)
**Statistical Inference:** Generalized Linear models, Survival Analysis, Hypothesis testing
**Version Control:** Git and GitHub
**Databases and Tools:** DepMap, UCSC Genome Browser, TCGA, ENCODE, NCBI, Ensembl, COSMIC
**Libraries:** PyTorch Ecosystem, Scikit-learn, matplotlib, NumPy, Pandas, ggplot2, dplyr, Shiny

## INDUSTRY EXPERIENCE

**Data Scientist - Grail Inc.** June 2022 - March 2025

- Developed & engineered a **Binomial Mixture Model** for early detection of **urothelial carcinoma** from **urine cell-free DNA**. [ASCO GU 2025]
  - Lead study design and data interpretation collaborations with wet-lab scientists and academic partners.
  - Designed and engineered classifier **hyperparameter tuning**, **calibration**, and **sensitivity analysis** experiments for detecting urothelial carcinoma from urine cell-free DNA.
  - Demonstrated 84% sensitivity at 95% specificity in an independent holdout set.
  - Designed and implemented **analytical validation studies** to assess **limit of detection (LoD)** and using dilution series and probabilistic modeling (probit regression) for ctDNA-based classifiers.
  - Applied classifier to revenue-generating **pharmaceutical partnerships**.
- Demonstrated that **urine cell-free DNA** improves **bladder cancer** detection compared to blood-based screening. [ASCO GU 2023]
  - Optimized tumor fraction estimation **Poisson Mixture Model** for urine cell-free DNA analyte, increasing accuracy & model robustness.
  - Demonstrated concordance between tissue-free and tissue-informed tumor fraction estimations.
- Contributed to the development of Grail's targeted methylation Multi-cancer Early Detection test.
  - Designed and **engineered data pipelines** and identified optimal training sets for the classifier intended for the latest release of **Galleri** (Grail's MCED product).
  - Lead **multi-modal analysis** efforts with Data Management and Clinical Data teams to integrate patient data it into the classifier.
- Developed and analyzed **validation studies** supporting **regulatory submissions** for oncology product development.

**Data Scientist Intern - Genentech** June 2021 - Sep 2021
Director: Dr. Matthew Grimmer

- Implemented a **Graph Concolutional Neural Network** to analyze cancer signatures and cancer pathways using the **DepMap** multi-omics dataset.

**Data Scientist Intern - Regeneron Pharmaceutical** June 2020 - Sep 2020
Director: Gregory Lallos

- Designed an end-to-end **image segmentation** pipeline to automatically detect cells from microscope images, replacing qualitative manual assessments.

**Data Scientist Intern - Agilent Technologies** June 2018 - Sep 2018
Director: Dr. Doug Roberts

- Performed anomaly detection in target enrichment experiments affecting DNA hybridization variability.
  - Designed novel DNA capturing probs using **Random Forest** which improved on-target coverage by 5%.
  - **Ensemblized** various machine learning classification models to enhance oligonucleotide classification.

**Data Scientist Intern - University of California, San Francisco** May 2016 - Sep 2016
Director: Prof. Sourav Bandypadhyay

- Developed an image classifier to identify different phases of the cell cycle using `CellProfiler`.

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## EDUCATION

**University of California, Santa Cruz**
**Ph.D., Bioinformatics** - Advisor: Prof. Josh Stuart Sep 2016-2022
*Developed an end to end pipeline to infer Gene Regulatory Networks in single-cells through integration of single-cell RNA-seq and single-cell ATAC-seq via **Multi-View Embedding** (e.g. CCA).*

- Benchmarked the pipeline using well-known PBMC regulons.
- Applied the pipeline to fetal heart data to identify regulatory elements across the developmental trajectory.

*Demonstrated that inflammation drives alternative first exon usage to regulate immune genes, including a novel iron-regulated isoform of Aim2 [eLife, (2021)]*

- Processed Nanopore and Illumina reads by performing **quality control** (FastQC), and aligning to the reference genome (**minimap2** for Nanopore, **TopHat** for Illumina).
- Performed differential gene expression analysis between LPS-stimulated and control macrophages

*In collaboration with Enara Health inc. conducted a retrospective **RWE** analysis using **Electronic Health Record (EHR)**-integrated clinical data to evaluate **digital health**-enabled obesity interventions.*

- Performed statistical analysis and demonstrated that medication timing and duration play a significant role in the weight-loss outcomes [*Obesity Science and Practice*, (2019)]
- Demonstrated the significant association of self-monitoring and consistent provider communication through the Enara mobile application (mHealth) with weight management[*Journal of Obesity and Chronic Diseases*, (2020)].

**University of California, Berkeley**
**B.A. Molecular and Cell Biology** - Advisor: Prof. Polina Lishko Aug 2014-2016

- Identified a novel lipid hydrolase progesterone receptor ABHD2, responsible for activating the principal calcium channel of spermatozoa and male fertilization [see *Science* Vol. 352, Issue 6285 - 2016].

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## PUBLICATIONS/POSTERS

- **Detection of early-stage urothelial cancers using methylation patterns in urine cell-free DNA**. Stewart, T., Shenoy, A., **Safavi, R.**, Stuart, S. M., McClintock, K., Alchaar, A., Bagrodia, A., Kader, K., McKay, R. R., Recio, N., Wagner, H., Fleshner, N. E., Larson, M. H., Salmasi, A., *ASCO GU, Feb 2025*
- **Single-cell chromatin accessibility reveals malignant regulatory programs in primary human cancers.**., *Science, Sep 2024*
- **Urine cell-free DNA improves detection of bladder cancer compared to blood-based screening**. Acknowledged for biopsy-informed TAF estimates - *Biopsy-informed TAF estimates generated by **Rojin Safavi**. ASCO GU, Feb 2023*
- **Impact of anti-obesity medication initiation and duration on weight loss in a comprehensive weight loss program**. **Safavi, R**., Lih, A., Kirkpatrick, S., Haller, S., and Bailony, M. R., *Obesity Science & Practice, July 2019*
- **Mobile Engagement Behaviors Associated with Sustained 15% Weight Loss**. Lih A, **Safavi R**, Isaq Z, Haller S, Bailony M. 2020, *Journal of Obesity and Chronic Diseases*

- **Inflammation Drives Alternative First Exon usage to Regulate Immune Genes including a Novel Iron Regulated Isoform of Aim2.** Elektra K. Robinson, Pratibha Jagannatha, Sergio Covarrubias, Matthew Cattle, **Rojin Safavi**, Ran Song, et al. - *eLife* 2021; DOI: 10.7554/eLife.69431
- **Prioritizing transcriptional factors in gene regulatory networks with PageRank.** Hongxu Ding, Ying Yang, Yuanqing Xue, Lucas Seninge, Henry Gong, Rojin Safavi et al., *iScience*, Volume 24, Issue 1, 2021
- **Unconventional Endocannabinoid Signaling Governs Sperm Activation via The Sex Hormone Progesterone**. Melissa R. Miller, Nadja Mannowetz, Anthony T. Iavarone, **Rojin Safavi**, Elena O. Gracheva, James F. Smith, Rose Z. Hill, Diana M. Bautista, Yuriy Kirichok, Polina V. Lishko, *Science*, 29 Apr 2016: Vol. 352, Issue 6285, pp. 555-559

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## AWARDS

Full **Regent's Fellowship** from UC Santa Cruz Program in Bioinformatics
Full **Fellowship** from UC Irvine Pharmaceutical Program Bioinformatics